DESIGN OPTIMIZATIN OF THE INSULATION DOMAIN IN A BUILDING WALL

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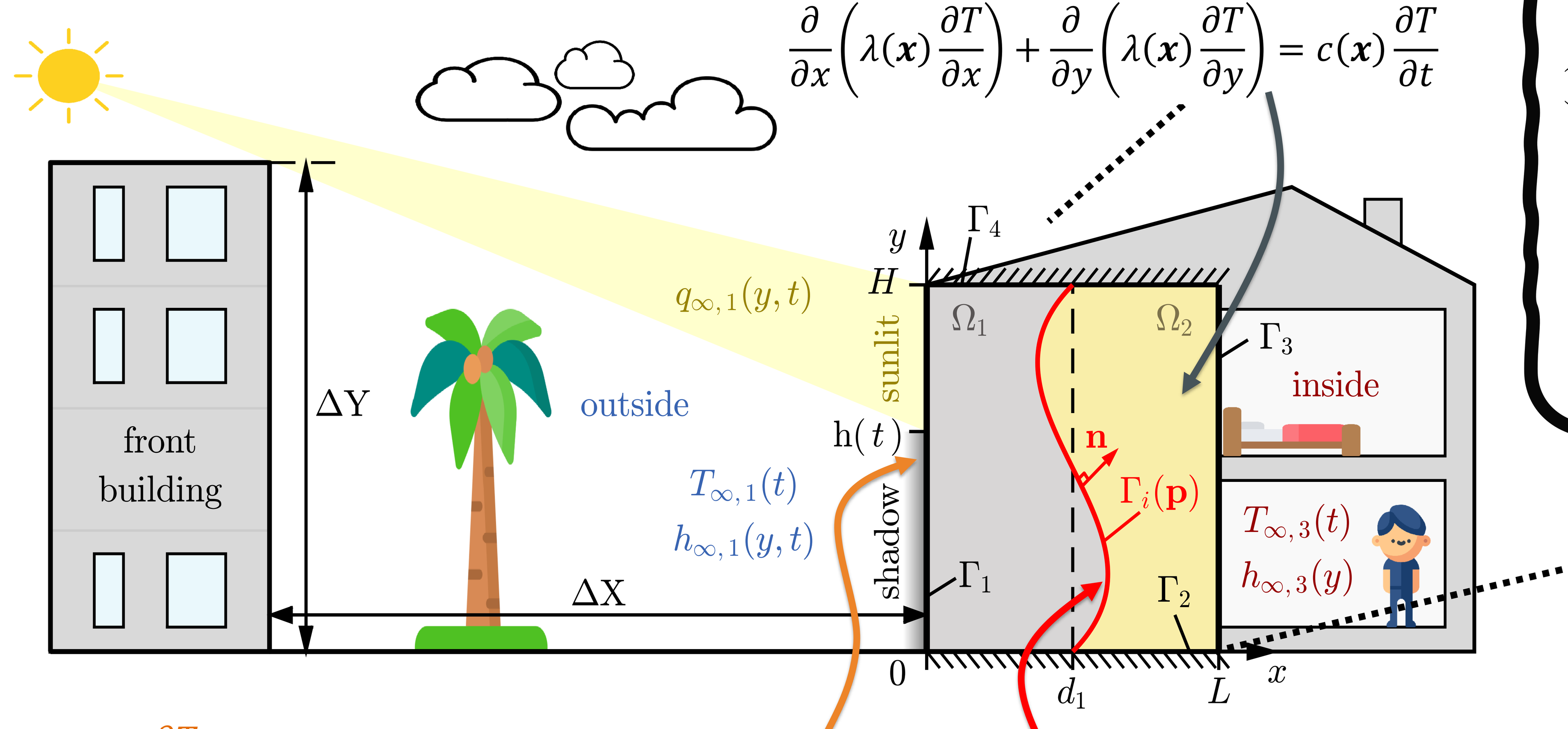
Introduction

- Building envelopes are subject to complex thermal loads: solar irradiation, wind-dependent convective coefficients.
- Conventional models rely on simplifying assumptions [1], which limit their ability to capture these boundary heterogeneities.
- **New opportunities:** additive manufacturing in construction enable complex geometries [2].
- **Research gap:** optimization studies mostly focus on steady-state analyses [3], neglecting transient environmental conditions.

Objective: This work presents a novel optimization framework for thermal wall design by geometrically optimizing the interface between structural and insulating materials. The objective is to minimize energy demand.

Physical model

$$\frac{\partial}{\partial x}\left(\lambda(\boldsymbol{x})\frac{\partial T}{\partial x}\right)+\frac{\partial}{\partial y}\left(\lambda(\boldsymbol{x})\frac{\partial T}{\partial y}\right)=c(\boldsymbol{x})\frac{\partial T}{\partial t}$$

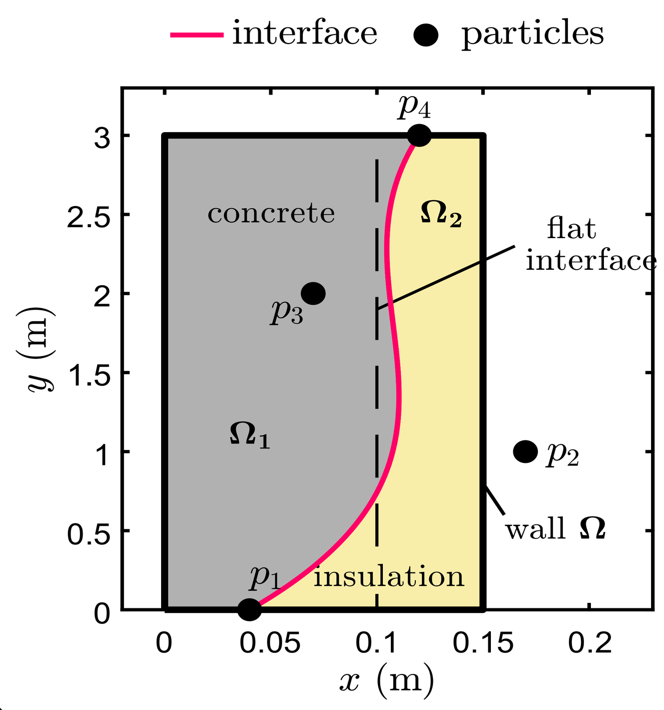


$$-\lambda(x)\frac{\partial T}{\partial x}=h_{\infty,1}(y,t)\left(T_{\infty,1}(t)-T\right)+q_{\infty,1}(y,t)$$

$$T(\boldsymbol{x}-\varepsilon)=T(\boldsymbol{x}+\varepsilon)$$

$$-\lambda(\boldsymbol{x})\left.\frac{\partial T}{\partial \mathbf{n}}\right|_{x-\varepsilon}=\lambda(\boldsymbol{x})\left.\frac{\partial T}{\partial \mathbf{n}}\right|_{x+\varepsilon}$$

Interface definition



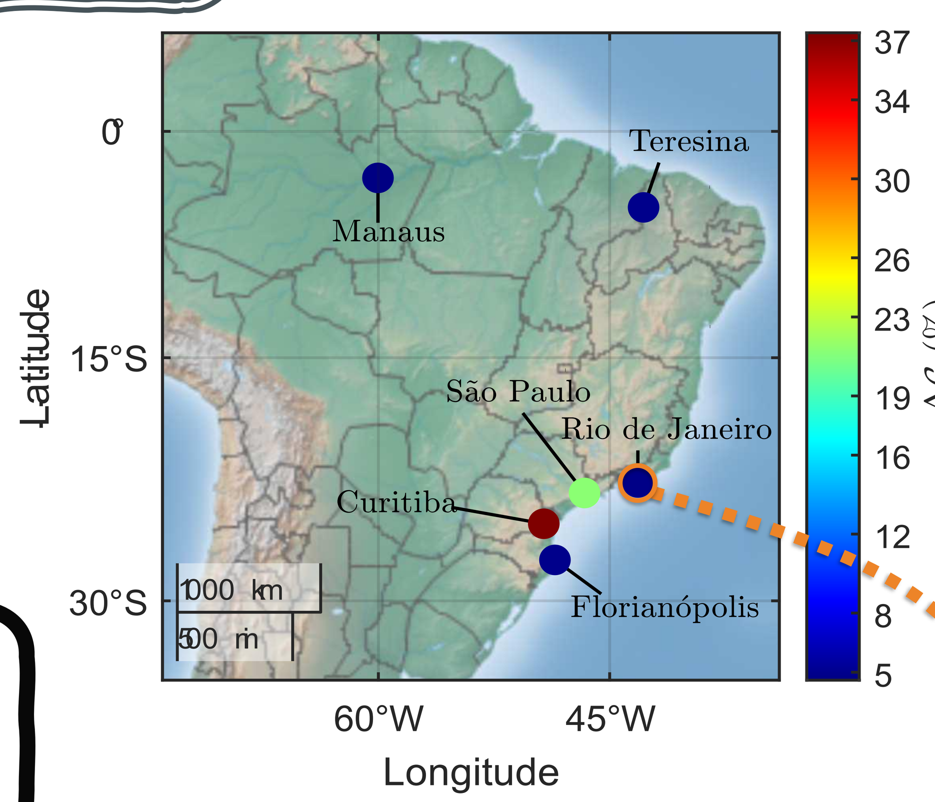
The interface Γ_i is represented using a Bézier curve defined with 4 control points and Bernstein polynomial interpolation:

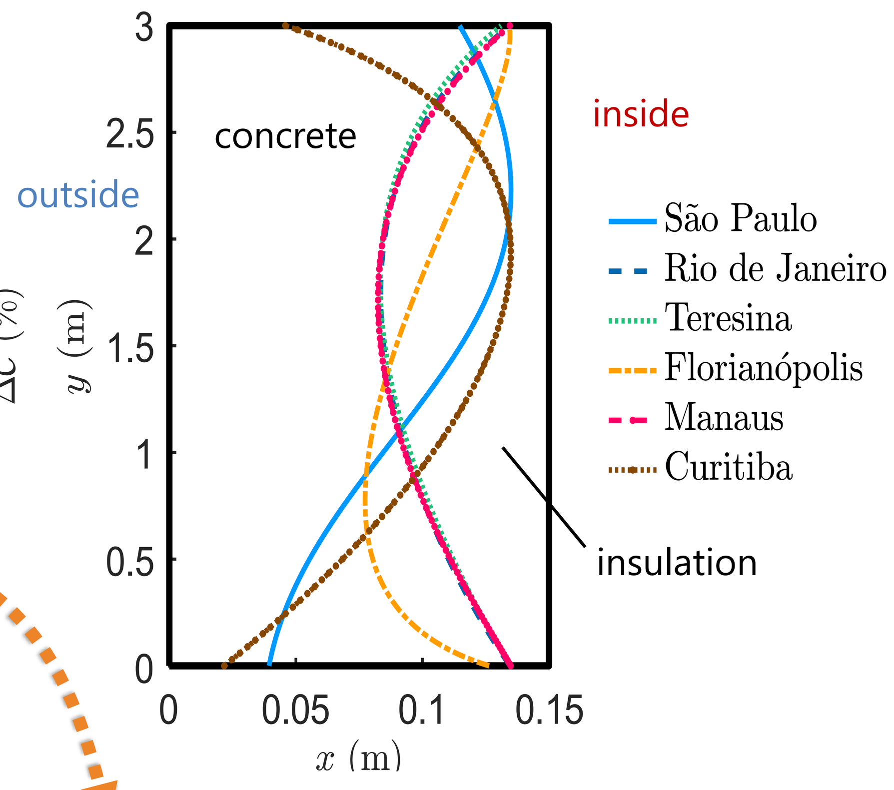
$$f_{B_x}(s,\mathbf{p})=\sum_{i=1}^{\mathrm{N}_\bullet}\mathcal{B}_{i-1}^{\mathrm{N}_\bullet}(s)\cdot x_i+\mathrm{A}(\mathbf{p})$$

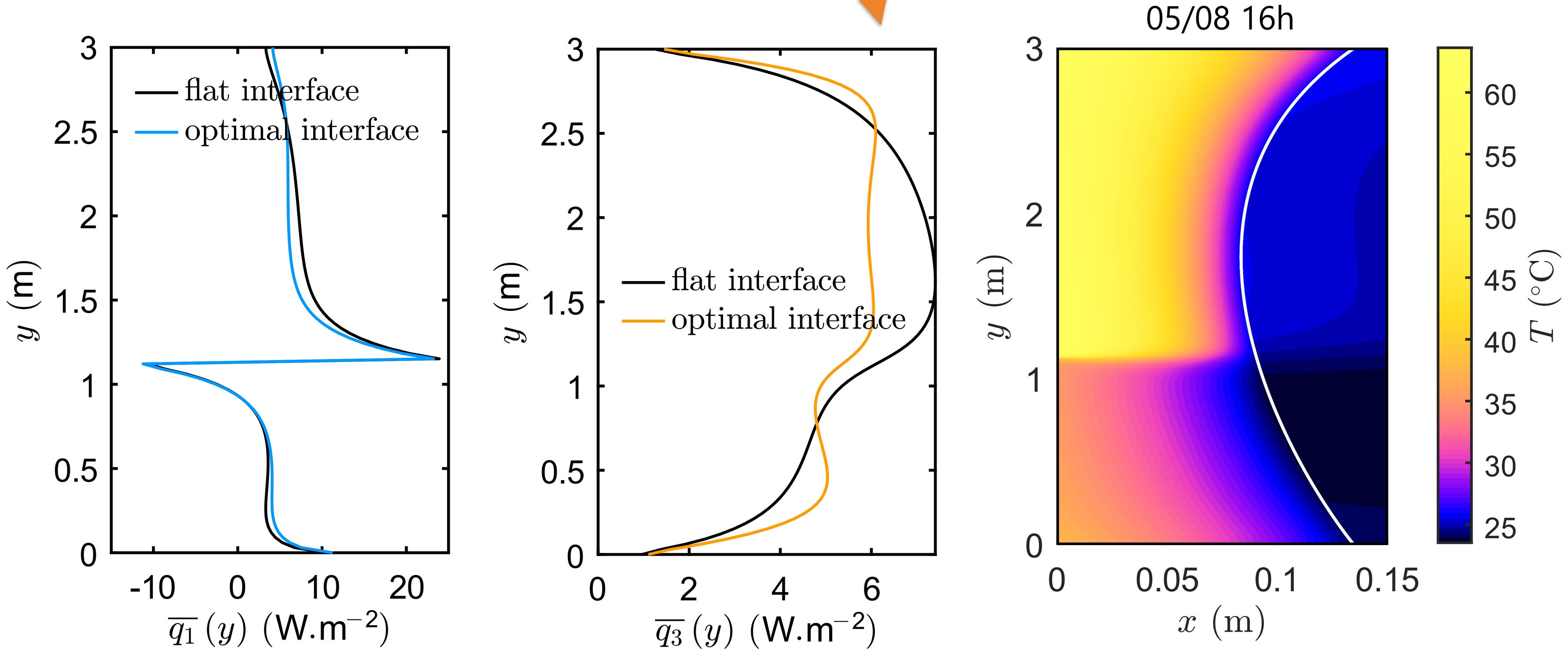
$$f_{B_y}(s,\mathbf{p})=\sum_{i=1}^{\mathrm{N}_\bullet}\mathcal{B}_{i-1}^{\mathrm{N}_\bullet}(s)\cdot y_i\,,\quad \forall s\in[0,1]$$

Case study in Brazil

- Promising performance tested under conditions of low internal insulation.
- The optimal interface in Rio de Janeiro redistributes and reduces the average internal heat flux $\overline{q_3}$ across the entire height of the wall, and reduces the magnitude of the average external heat flux $\overline{q_1}$ compared to a flat interface.







Thermal design problem

The minimization problem aims to find the optimal coordinates of the control points $\boldsymbol{p}^\star=\min_{\boldsymbol{p}\in\Omega_p}\mathcal{C}$ of the Bézier curve, defined as:

$$\mathcal{C}=\int_0^{t_f}w(t)\,\mathrm{d}t\,,\ \text{with}\ \ w(t)=\left(\int_{\Gamma_3}-\lambda(\mathbf{x})\cdot\frac{\partial T}{\partial x}\,\mathrm{d}y\right)\cdot\left(\frac{T_{\infty,1}(t)}{T_{\infty,3}(t)}-1\right)$$

subject to:
- Volume constraint,
- Spatial constraint.

NSGA-II convergence assessment = relative objective variation $\delta\mathcal{C}$ + mean Euclidean parameter dispersion

Poster is animated !

Conclusion & perspectives

Despite geometric limitations (Bézier curves with $\mathrm{N}_\bullet = 4$ control points) and the absence of building-scale thermal simulations, annual energy gains of up to nearly 40% are achieved for the north-oriented façade compared with flat interfaces, highlighting the **potential of interface geometry optimization**.

References

[1] M. Woloszyn and C. Rode. *Tools for performance simulation of heat, air and moisture conditions of whole buildings*. Building Simulation, 2008.
[2] G. Vantyghem et al., *Density-based topology optimization for 3D-printable building structures*, Structural and Multidisciplinary Optimization, 2019.
[3] Z. Karashbayeva et al., *Optimization of the insulation domain for heat transfer in building walls*, Journal of the Brazilian Society of Mechanical Sciences and Engineering, 2025.

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